

4 Challenges and opportunities

Enquiry question: *What challenges and opportunities exist in cold environments and what management issues might result from their use?*

What students need to learn	Suggested teaching and learning
■ The terms challenges and opportunities and explore the link between them.	■ Developing an awareness of the terms challenges and opportunities and understanding the link between them.
■ The challenges and opportunities of cold environments, including those of the present and the past.	■ Researching the wide range of challenges (relief, climate, avalanches, surges, meltwater floods, ground conditions, fragile ecosystems etc) and opportunities (tourism, HEP, water supply, mineral resources, etc) in different cold environments.
■ Cold environments in which humans are attempting to take the opportunities available by overcoming the challenges. These should include both glacial and periglacial environments.	■ Investigating how changes in demand and technology could lead to challenges being overcome from examples of a range of cold environments.
■ The effectiveness of different approaches to using and managing cold environments, considering the attitudes of different groups (residents, government, pressure groups, business/industry) and the conflicts that can exist between them.	■ Researching the values and attitudes of the different interest groups involved in the use and management of cold environments and examining the different approaches that can be taken to management; such as protection/conservation and sustainable management, should it be deemed necessary.

Suggested fieldwork opportunities for cold environments – landscape and change

Local fieldwork

- Upland fieldwork to investigate glacial processes and the past distribution of glacial landscapes in the UK.
- Upland alpine environments could be investigated in terms of fragility and threats.

Residential and long-haul fieldwork

- Both the Alps and Pyrennes afford good opportunities for the study of the full range of a processes. Norway and Iceland would also be excellent locations for extended fieldwork.

4.5 Option 3: Life on the Margins: the Food Supply Problem

Life on the margins Significant numbers of people live a life on the margin, in a situation of food insecurity. Others consume more than their fair share of global resources. This option explores this inequality, focusing on regions where food production is a continual challenge. The 'margins' may be traditional areas of famine, but also rapidly urbanising areas where food is scarce and malnutrition an ever present threat.

Food insecurity The causes of food insecurity are complex, ranging from physical processes of land degradation and desertification, exacerbated by human over-exploitation, to population pressure and political processes. Increasing food supply represents a key challenge cemented with the Millennium Development Goals. However, often agreement does not exist on the best way to achieve this; options range from reforming trade systems to hi-tech farming, intermediate technology and organic farming.

Global synoptic context

Key linkages

Places

Where are the 'margins'?

People

Who suffers from food insecurity and over-nutrition?

Power

Which organisations and groups are involved in ensuring food security?

Suggested teaching and learning

Mapping the distribution of countries suffering from food insecurity, and those facing the obesity threat.

Contrasting the socio-economic profiles of countries at different end of the food supply spectrum.

Briefly investigating the role of UN bodies such as the FAO, and NGOs such as Practical Action.

1 Global and local feast or famine

Enquiry question: *What are the characteristics of food supply and security?*

What students need to learn	Suggested teaching and learning
■ Current issues associated with food supply and security; such as food miles, famine, globalisation of food tastes, under and over nutrition.	■ Exploring the current socio-economic issues associated with feeding people globally and locally.
■ The environmental issues resulting from food production.	■ Developing an awareness of inappropriate farming techniques and their impact on the environment.
■ Why food supply varies spatially.	■ Investigate how current patterns of food supply and security vary at different scales in rural and urban environments.
■ What life on the margins means to differing people and the role of food security in the overall quality of life there.	■ Developing an awareness that life on the margins of survival ranges from traditionally perceived areas of famine such as the Sudan to megacities where poverty has impacts on food supply and consumption.

2 The complex causes of food supply inequalities

Enquiry question: *What has caused global inequalities in food supply and security?*

What students need to learn	Suggested teaching and learning
■ The complex interlinking and overlapping causes of famine and food surpluses and classification of the environmental, social and economic causes into long- and short-term, direct and indirect causes.	■ Investigating how and why areas of food surplus and famine have emerged.
■ The role of population pressure in creating food insecurity.	■ Weighing up the contrasting viewpoints on population pressure as highlighted by neo-Malthusians versus technocentric followers of Boserup.
■ The impacts of attempts to increase global food supply.	■ Researching food production techniques which have been environmentally damaging and contrasting these with more environmentally friendly production methods such as LEAF and organic schemes in locations as diverse as the UK, Cuba, China and Africa.
■ Who has been most affected by food insecurity, and why.	■ Investigating the nature of groups of people vulnerable to food insecurity.

3 Desertification and life at the margin of survival

Enquiry question: *What is the role of desertification in threatening life at the margins?*

What students need to learn	Suggested teaching and learning
■ Desertification and its scale and impact.	■ Developing an awareness that desertification is a type of land degradation and investigating the varying human activities and climatic variations causing it.